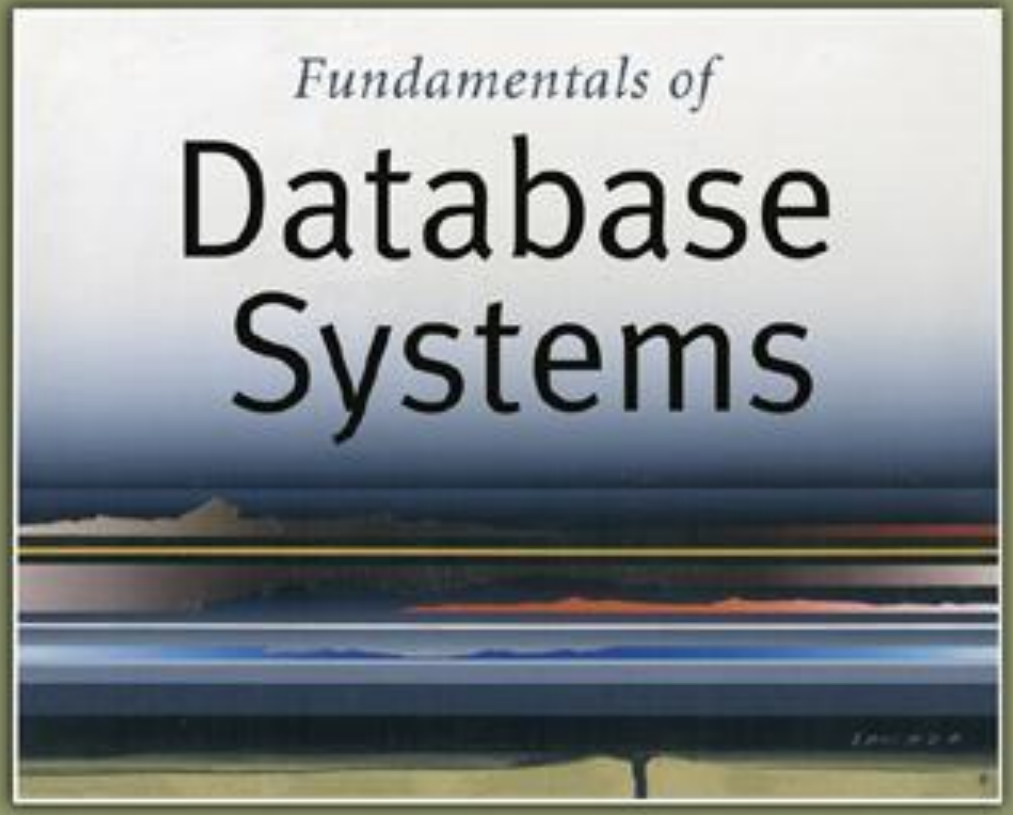

5th Edition

Elmasri / Navathe

Module 2

Enhanced Entity-Relationship (EER) Modeling

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Chapter Outline

- EER stands for Enhanced ER or Extended ER
- EER Model Concepts
  - Includes all modeling concepts of basic ER
  - Additional concepts:
    - subclasses/superclasses
    - specialization/generalization
    - categories (UNION types)
    - attribute and relationship inheritance
  - These are fundamental to conceptual modeling
- The additional EER concepts are used to model applications more completely and more accurately
  - EER includes some object-oriented concepts, such as inheritance

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Enhanced Entity Relationship model

It is a diagrammatic technique for displaying the following concepts:

- Sub Class and Super Class
- Specialization and Generalization
- Union or Category
- Aggregation

Subclasses and Superclasses

- An entity type may have additional meaningful subgroupings of its entities
  - Example: EMPLOYEE may be further grouped into:
    - SECRETARY, ENGINEER, TECHNICIAN, …
      - Based on the EMPLOYEE's Job
    - MANAGER
      - EMPLOYEEs who are managers
    - SALARIED_EMPLOYEE, HOURLY_EMPLOYEE
      - Based on the EMPLOYEE's method of pay
- EER diagrams extend ER diagrams to represent these additional subgroupings, called *subclasses* or *subtypes*

- Sub class and Super class relationship leads the concept of Inheritance.

- **1. Super Class:** Super class is an entity type that has a relationship with one or more subtypes.

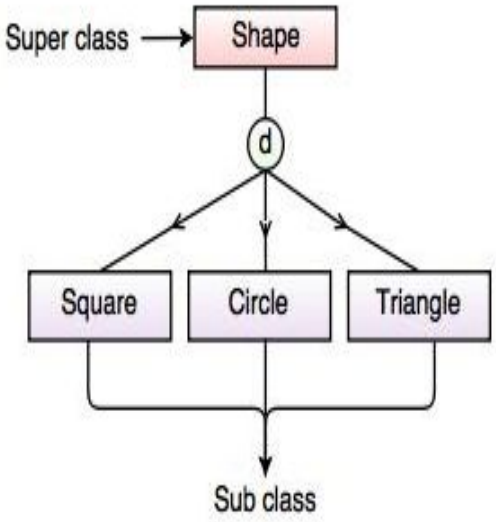
For example: Shape super class is having sub groups as Squ

- **2. Sub Class: It** is a group of entities with unique
- attributes.
- Sub class inherits properties and attributes

From its super class.

For example: Square, Circle, Triangle sub class

of Shape super class.



Subclasses and Superclasses (3)

- These are also called IS-A relationships
  - SECRETARY IS-A EMPLOYEE, TECHNICIAN IS-A EMPLOYEE, ….
- Note: An entity that is member of a subclass represents the same real-world entity as some member of the superclass:
  - The subclass member is the same entity in a *distinct specific role*
  - An entity cannot exist in the database merely by being a member of a subclass; it must also be a member of the superclass
  - A member of the superclass can be optionally included as a member of any number of its subclasses

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Subclasses and Superclasses (4)

- Examples:
  - A salaried employee who is also an engineer belongs to the two subclasses:
    - ENGINEER, and
    - SALARIED_EMPLOYEE
  - A salaried employee who is also an engineering manager belongs to the three subclasses:
    - MANAGER,
    - ENGINEER, and
    - SALARIED_EMPLOYEE
- It is not necessary that every entity in a superclass be a member of some subclass

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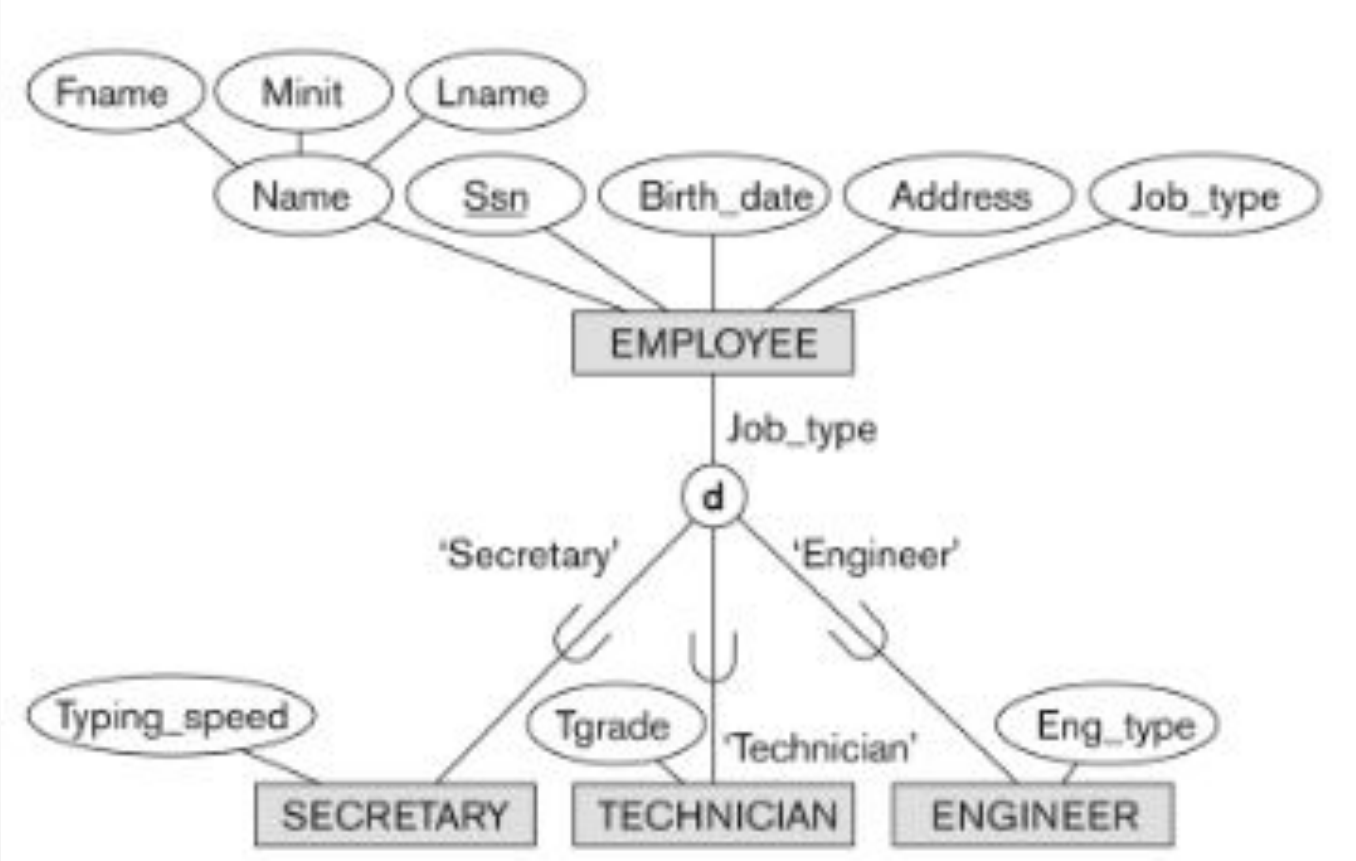
Representing Specialization in EER Diagrams



Attribute Inheritance in Superclass / Subclass Relationships

- An entity that is member of a subclass *inherits*
  - All attributes of the entity as a member of the superclass
  - All relationships of the entity as a member of the superclass
- Example:
  - In the previous slide, SECRETARY (as well as TECHNICIAN and ENGINEER) inherit the attributes Name, SSN, …, from EMPLOYEE
  - Every SECRETARY entity will have values for the inherited attributes

Specialization (1)

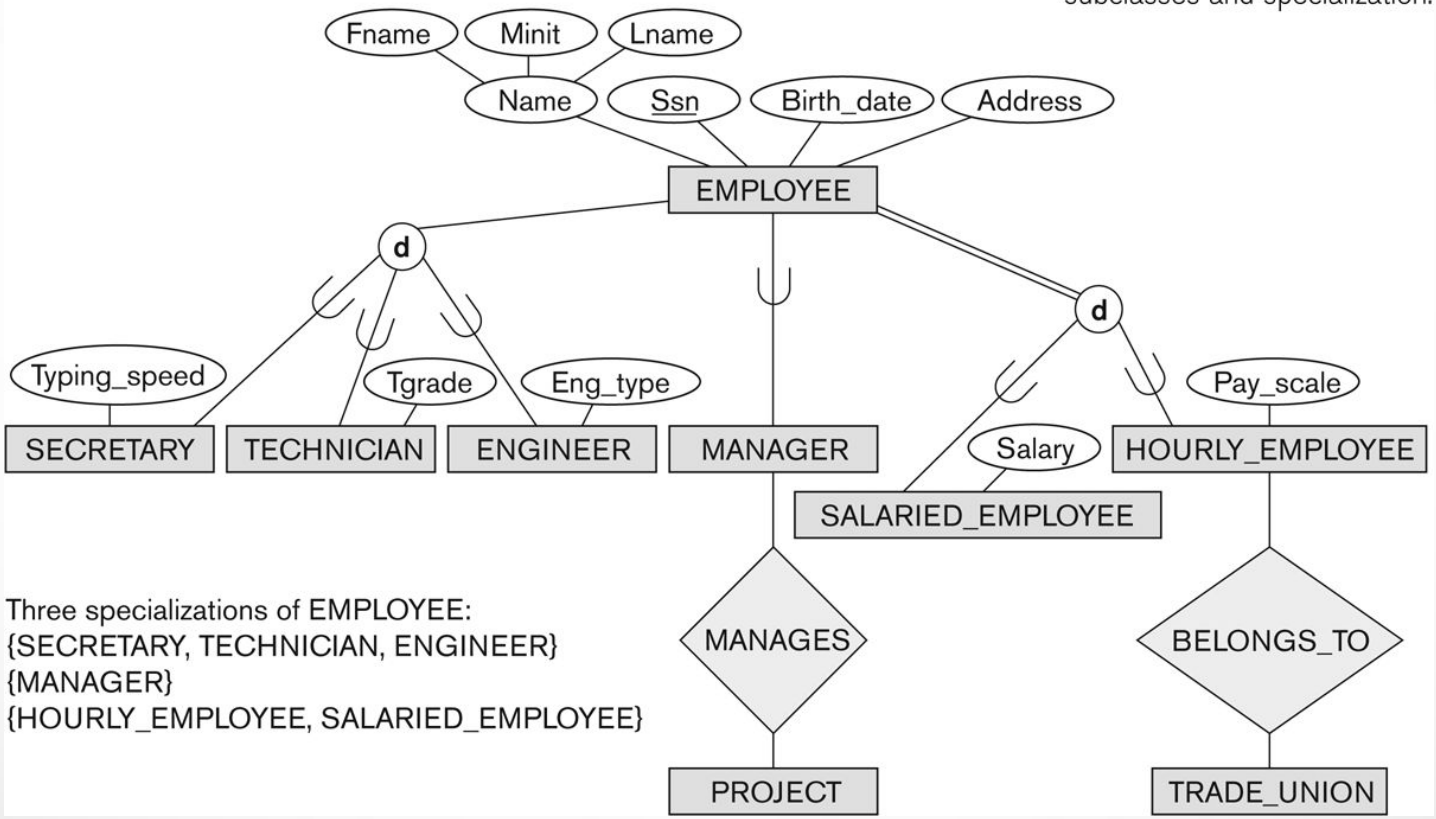
- Specialization is the process of defining a set of subclasses of a superclass
- The set of subclasses is based upon some distinguishing characteristics of the entities in the superclass
  - Example: {SECRETARY, ENGINEER, TECHNICIAN} is a specialization of EMPLOYEE based upon *job type.*
    - May have several specializations of the same superclass

Specialization (2)

- Example: Another specialization of EMPLOYEE based on *method of pay* is {SALARIED_EMPLOYEE, HOURLY_EMPLOYEE}.
  - Superclass/subclass relationships and specialization can be diagrammatically represented in EER diagrams
  - Attributes of a subclass are called *specific* or *local* attributes.
    - For example, the attribute TypingSpeed of SECRETARY
  - The subclass can also participate in specific relationship types.
    - Now suppose: Only Permanent_Employee gets PF:
    - Permanent_Employee HAS_PF Provident_Fund

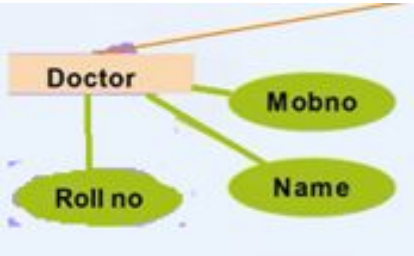
Specialization (3)

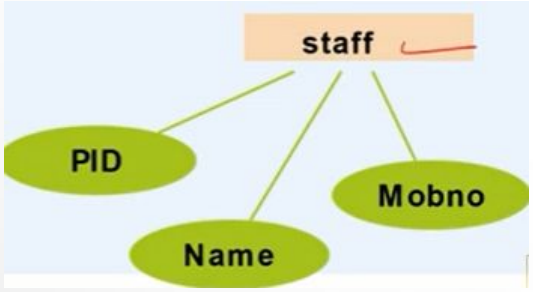
Figure 4.1
EER diagram notation to represent subclasses and specialization.

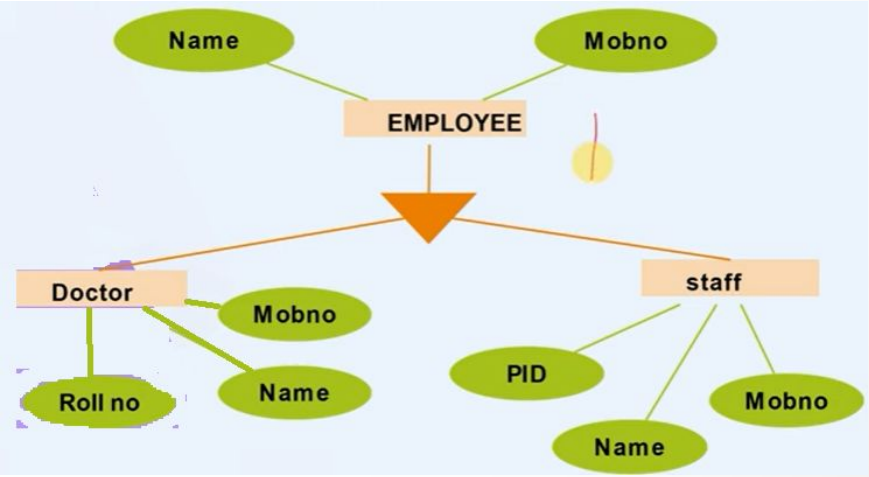


Three specializations of EMPLOYEE:
{SECRETARY, TECHNICIAN, ENGINEER}
{MANAGER}
{HOURLY_EMPLOYEE, SALARIED_EMPLOYEE}

Example of Specialization

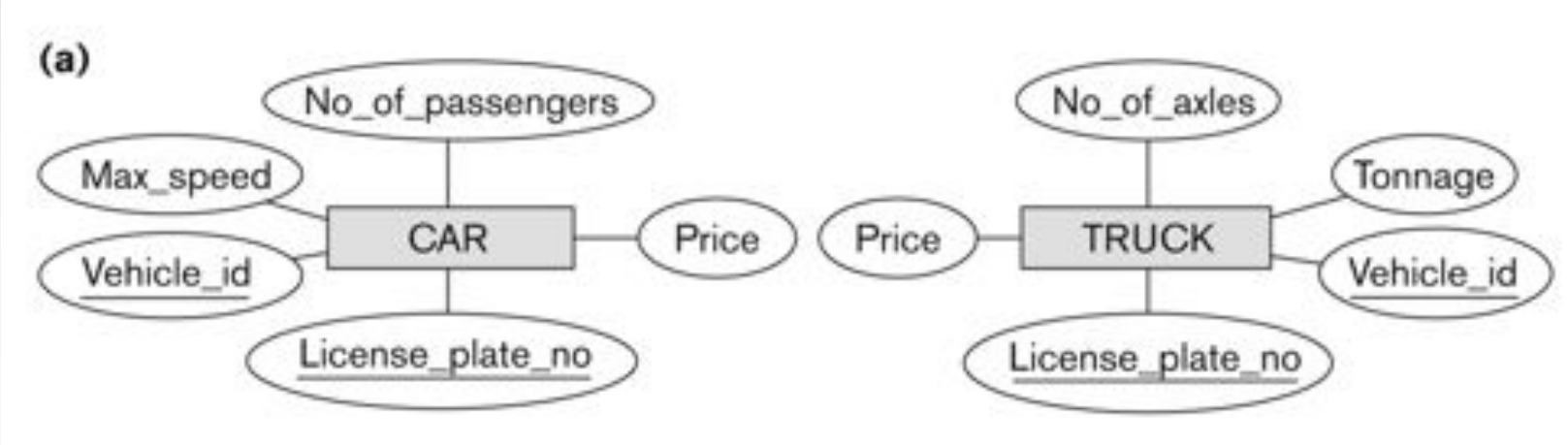


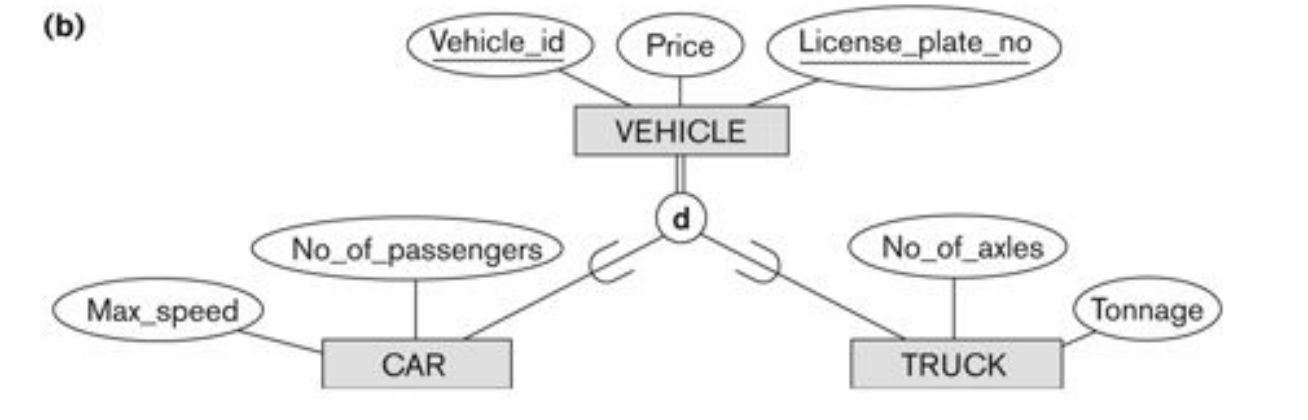




Generalization

- Generalization is the reverse of the specialization process
- Several classes with common features are generalized into a superclass;
  - original classes become its subclasses
- Example: CAR, TRUCK generalized into VEHICLE;
  - both CAR, TRUCK become subclasses of the superclass VEHICLE.
  - We can view {CAR, TRUCK} as a specialization of VEHICLE
  - Alternatively, we can view VEHICLE as a generalization of CAR and TRUCK





Slide 4- 16

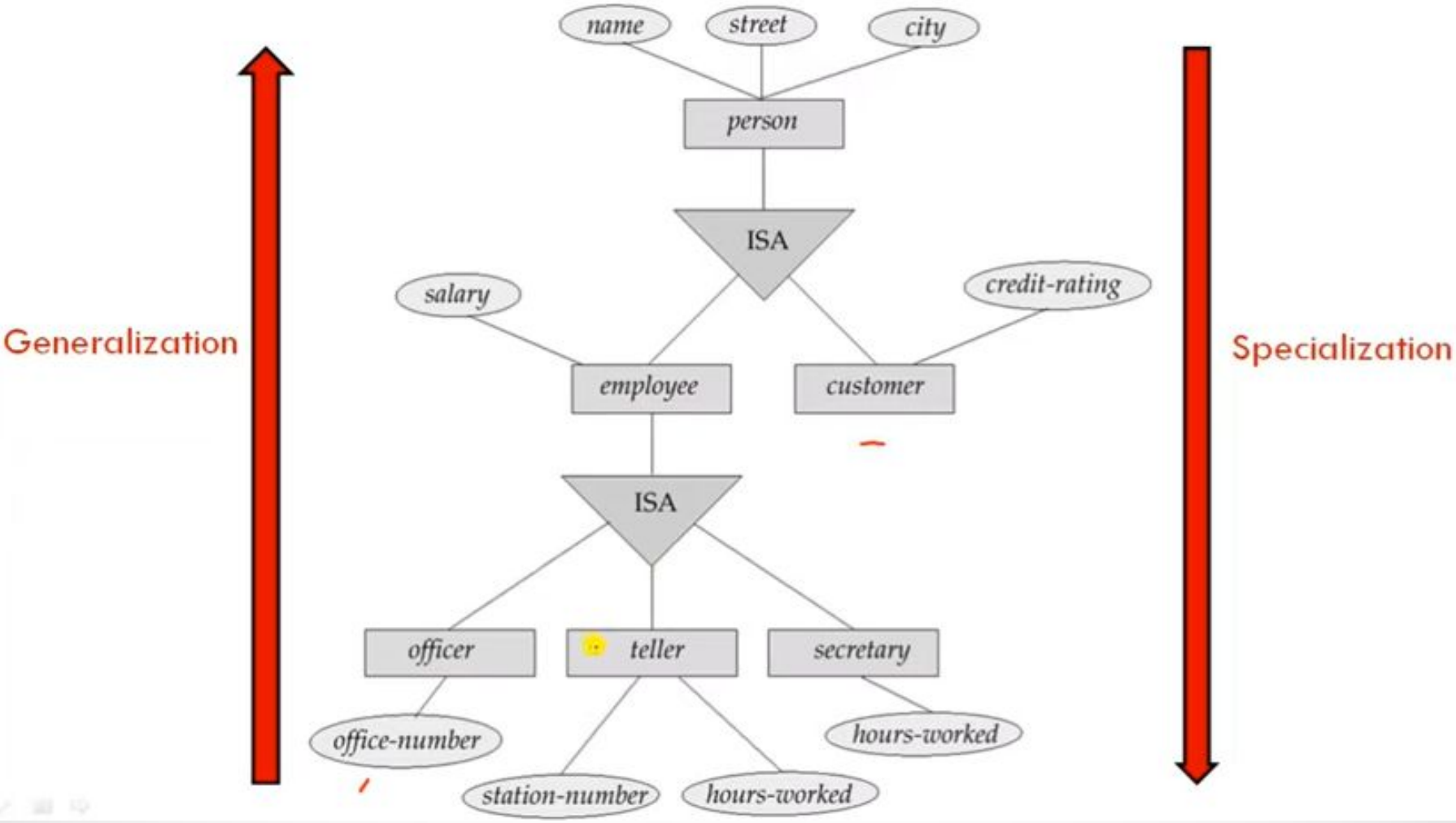
Generalization. (a) Two entity types, CAR and TRUCK. (b) Generalizing CAR and TRUCK into the superclass VEHICLE.

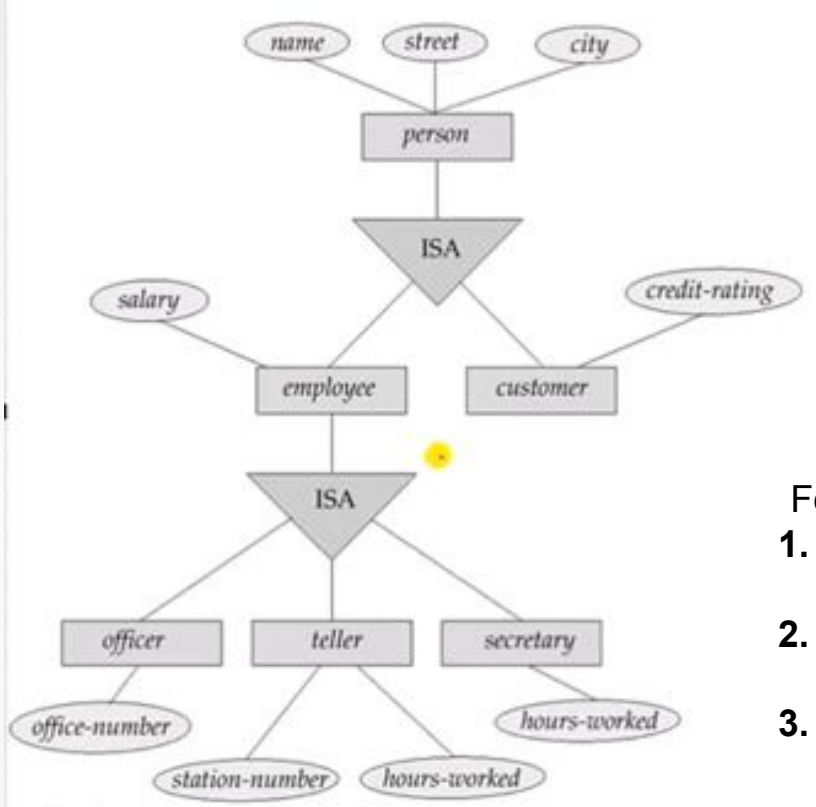
Generalization and Specialization (1)

- Diagrammatic notation are sometimes used to distinguish between generalization and specialization
  - Arrow pointing to the generalized superclass represents a generalization
  - Arrows pointing to the specialized subclasses represent a specialization
  - We *do not use* this notation because it is often subjective as to which process is more appropriate for a particular situation
  - We advocate not drawing any arrows

Generalization and Specialization (2)

- Data Modeling with Specialization and Generalization
  - A superclass or subclass represents a collection (or set or grouping) of entities
  - It also represents a particular *type of entity*
  - Shown in rectangles in EER diagrams (as are entity types)
  - We can call all entity types (and their corresponding collections) ***classes***, whether they are entity types, superclasses, or subclasses


Generalization
Specialization
name
street
city
person
ISA
salary
credit-rating
employee
customer
ISA
officer
teller
secretary
office-number
station-number
hours-worked
hours-worked



Four tables can be formed:

1. **Customer**(name,street,city,credit_rating)

2. **Officer**(name,street,city,salary,office_number)

3. **Teller**(name,street,city,salary,station_number,hours_worked)

4. **Secretary**(name,strret,city,hours_worked)

Constraints on Specialization and Generalization (1)

- If we can determine exactly those entities that will become members of each subclass by a condition, the subclasses are called predicate-defined (or condition-defined) subclasses
  - Condition is a constraint that determines subclass members
  - Display a predicate-defined subclass by writing the predicate condition next to the line attaching the subclass to its superclass

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Constraints on Specialization and Generalization (2)

- If all subclasses in a specialization have membership condition on same attribute of the superclass, specialization is called an attribute-defined specialization
  - Attribute is called the defining attribute of the specialization
  - Example: JobType is the defining attribute of the specialization {SECRETARY, TECHNICIAN, ENGINEER} of EMPLOYEE
- If no condition determines membership, the subclass is called user-defined
  - Membership in a subclass is determined by the database users by applying an operation to add an entity to the subclass
  - Membership in the subclass is specified individually for each entity in the superclass by the user

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Attribute-Defined Specialization

- The subclass is created based on the value of an attribute of the superclass.
- The system automatically decides the subclass based on the attribute value.
- No human intervention is needed to decide the membership.
- Example: Employee → Permanent_Employee / Hourly_Employee

Employee_ID	Name	Type
101	Ravi	Permanent
102	Sita	Hourly
103	John	Permanent

Here, the attribute Type determines the subclass.

If Type = Permanent, it automatically goes to Permanent_Employee.

If Type = Hourly, it automatically goes to Hourly_Employee.

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User-Defined Specialization

- The subclass is created based on a decision made by the user (designer/administrator), not automatically.
- The system does not have an attribute to decide membership.
- Membership is assigned manually or by a business rule.
- Example: Employee → Management_Committee / Sports_Team.

- Not every employee has an attribute that says “is in management committee” or “in sports team.”
- The administrator decides who belongs to which subclass.

- Subclass membership is manual and chosen by the user.

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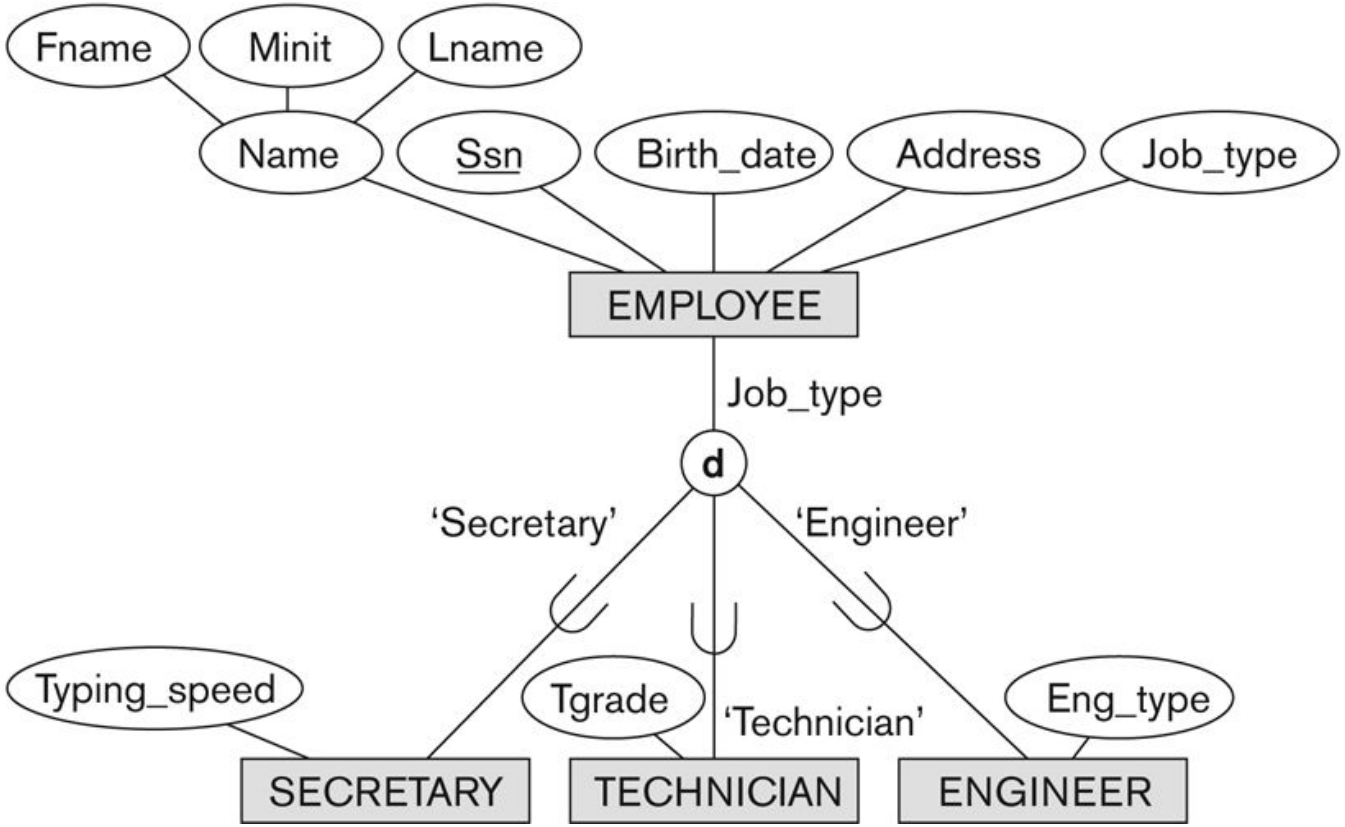
Displaying an attribute-defined specialization in EER diagrams

Figure 4.4
EER diagram notation for an attribute-defined specialization on Job_type.



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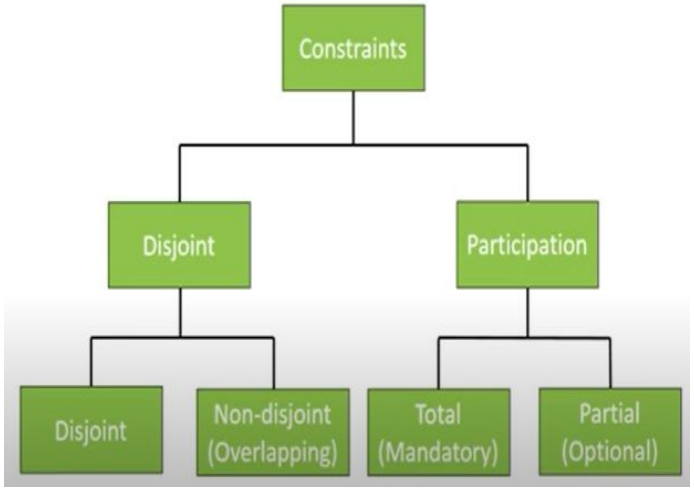
Constraints on Specialization and Generalization (3)

- Two basic constraints can apply to a specialization/generalization:
  - Disjointness Constraint:
  - Completeness/participation Constraint:
  - Attribute Constraints:



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Constraints on Specialization and Generalization (4)

- Disjointness Constraint:
  - Specifies that the subclasses of the specialization must be *disjoint*:
    - an entity can be a member of at most one of the subclasses of the specialization
  - Specified by ***d*** in EER diagram
  - If not disjoint, specialization is *overlapping*:
    - that is the same entity may be a member of more than one subclass of the specialization
  - Specified by ***o*** in EER diagram

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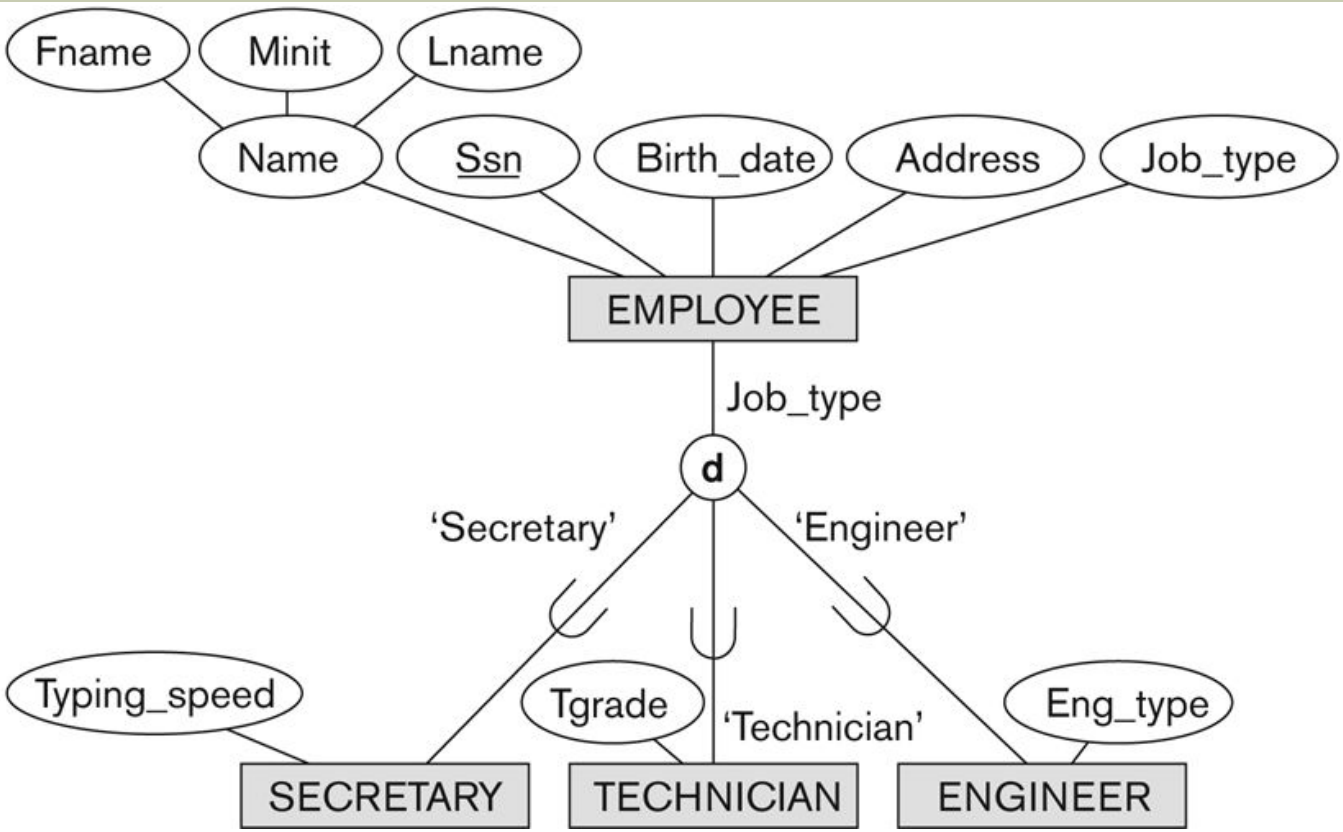
Example

Figure 4.4
EER diagram notation for an attribute-defined specialization on Job_type.



- In the diagram Employee can be either Secretary or Technician or Engineer

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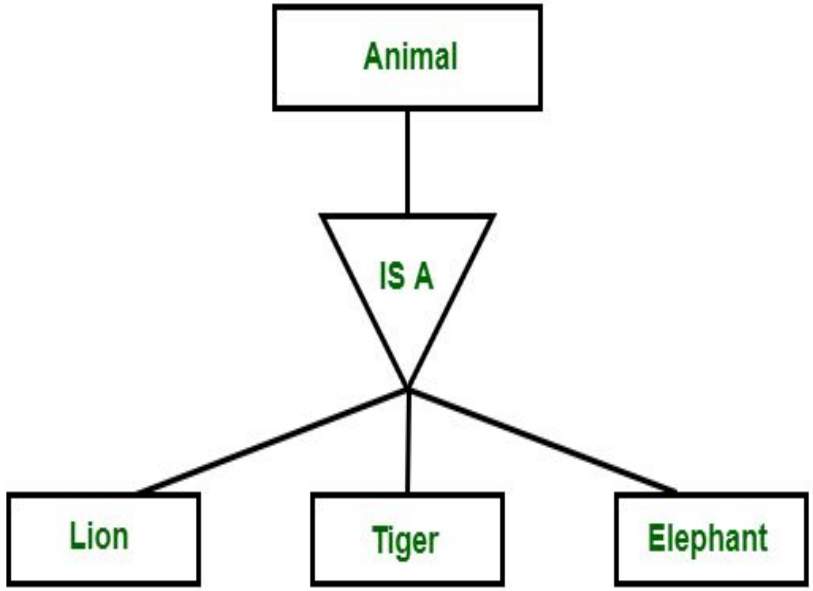
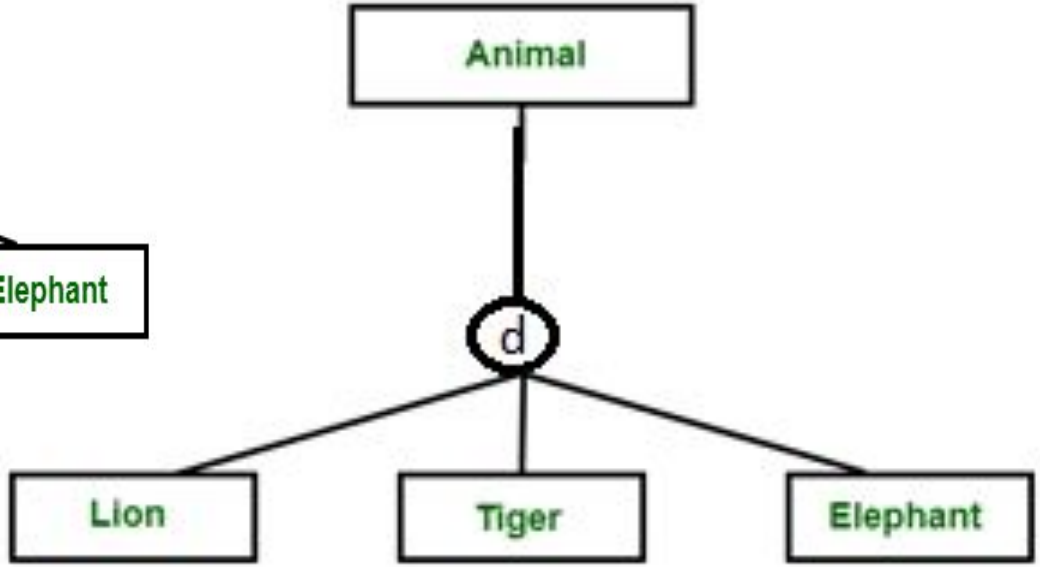
Disjoint Constraint





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Constraints on Specialization and Generalization (5)

- Completeness Constraint:
  - *Total* specifies that every entity in the superclass must be a member of some subclass in the specialization/generalization
  - Shown in EER diagrams by a ***double line***
  - *Partial* allows an entity not to belong to any of the subclasses
  - Shown in EER diagrams by a single line

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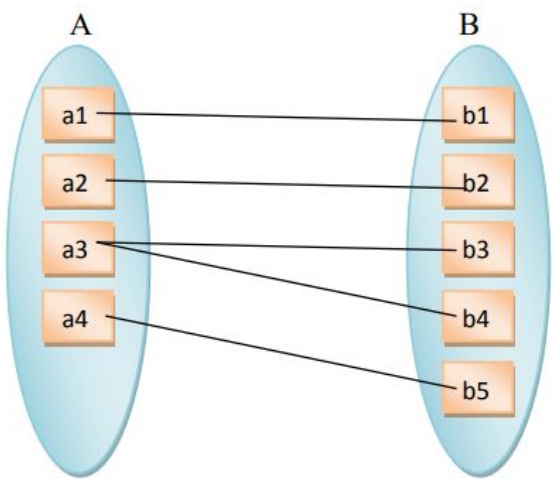
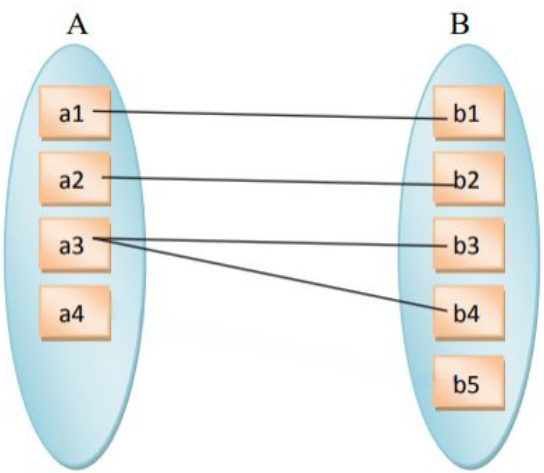
- Total participation Partial Participation





Participation in Enrolled relationship set: **Partial** Course
Total Student

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Constraints on Specialization and Generalization (6)

- Hence, we have four types of specialization/generalization:
  - Disjoint, total
  - Disjoint, partial
  - Overlapping, total
  - Overlapping, partial
- Note: Generalization usually is total because the superclass is derived from the subclasses.

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Disjoint, Total (⊥ , ⊆)

- Superclass: Employee
- Subclasses:
  - Permanent_Employee
  - Hourly_Employee
- Rules:
- An employee cannot be both permanent and hourly → Disjoint
- Every employee must be either permanent or hourly → Total

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Disjoint, Partial (⊥ , ⊄)

- Superclass: Vehicle
- Subclasses:
  - Car
  - Bike
- Rules:
- A vehicle cannot be both car and bike → Disjoint
- But there can be vehicles that are neither (e.g., Truck, Bus) → Partial

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Overlapping, Total (O, T)

- Superclass: Person
- Subclasses:
  - Student
  - Employee
- Rules:
- A person can be both student and employee → Overlapping
- Every person in this model must be at least one of these (student or employee) → Total

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Example of overlapping total Specialization

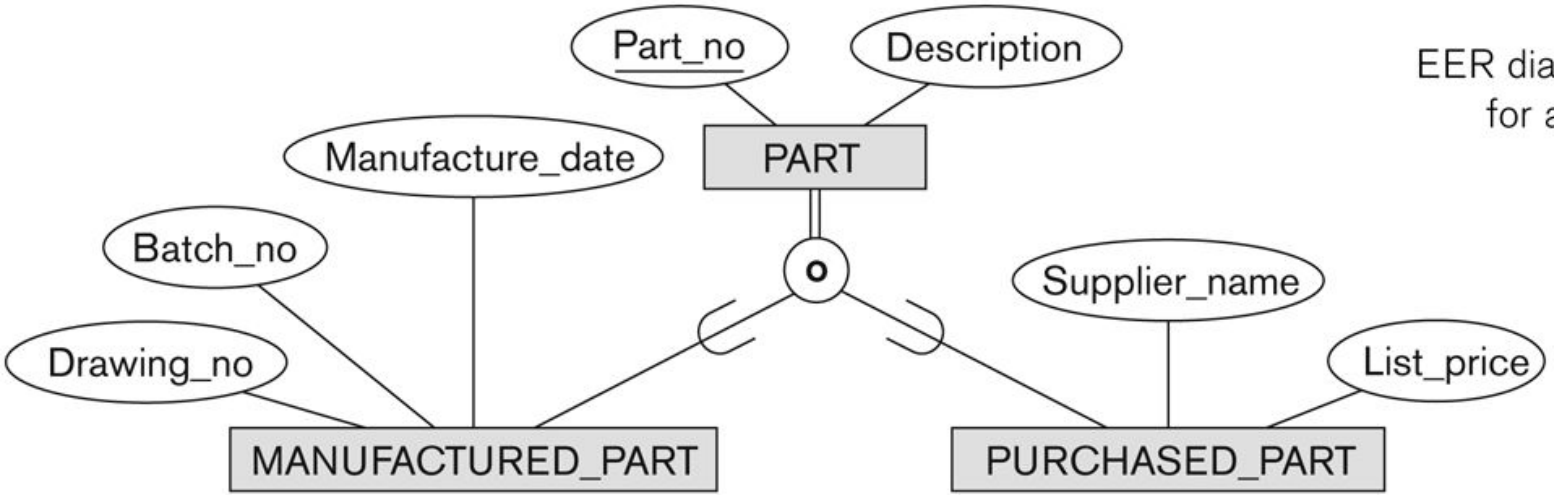

Figure 4.5
EER diagram notation for an overlapping (nondisjoint) specialization.

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Overlapping, Partial (O, P)

- Superclass: Person
- Subclasses:
  - Doctor
  - Researcher
- Rules:
- A person can be both doctor and researcher → Overlapping
- But some persons may be neither (e.g., farmer, artist) → Partial

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Specialization/Generalization Hierarchies, Lattices & Shared Subclasses (1)

- A subclass may itself have further subclasses specified on it
  - forms a hierarchy or a lattice
- ***Hierarchy*** has a constraint that every subclass has only one superclass (called ***single inheritance***); this is basically a ***tree structure***
- In a ***lattice***, a subclass can be subclass of more than one superclass (called ***multiple inheritance***)

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Specialization/Generalization Hierarchies, Lattices & Shared Subclasses (2)

- In a lattice or hierarchy, a subclass inherits attributes not only of its direct superclass, but also of all its predecessor superclasses
- A subclass with more than one superclass is called a **shared subclass** (multiple inheritance)
- Can have:
  - *specialization* hierarchies or lattices, or
  - *generalization* hierarchies or lattices,
  - depending on how they were *derived*
- We just use *specialization* (to stand for the end result of either specialization or generalization)

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Shared Subclass “Engineering_Manager”

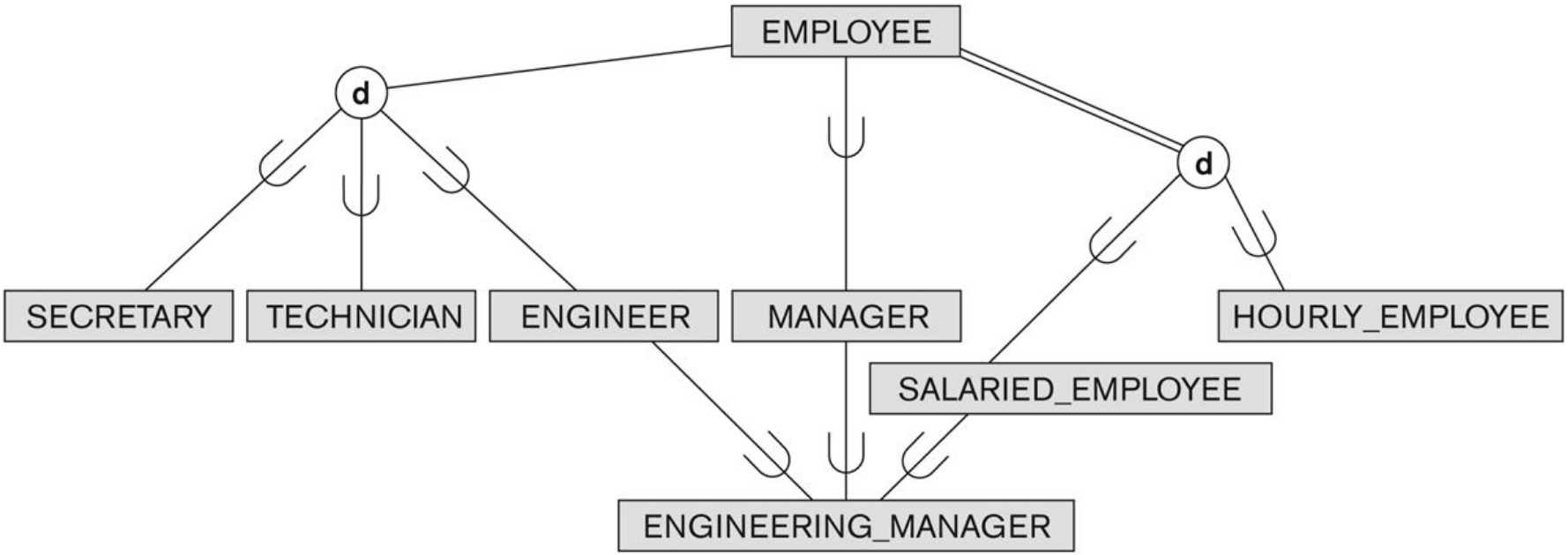

Figure 4.6
A specialization lattice with shared subclass ENGINEERING_MANAGER.

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Specialization/Generalization Hierarchies, Lattices & Shared Subclasses (3)

- In *specialization*, start with an entity type and then define subclasses of the entity type by successive specialization
  - called a *top down* conceptual refinement process
- In *generalization*, start with many entity types and generalize those that have common properties
  - Called a *bottom up* conceptual synthesis process
- In practice, a *combination of both processes* is usually employed

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Specialization / Generalization Lattice Example (UNIVERSITY)

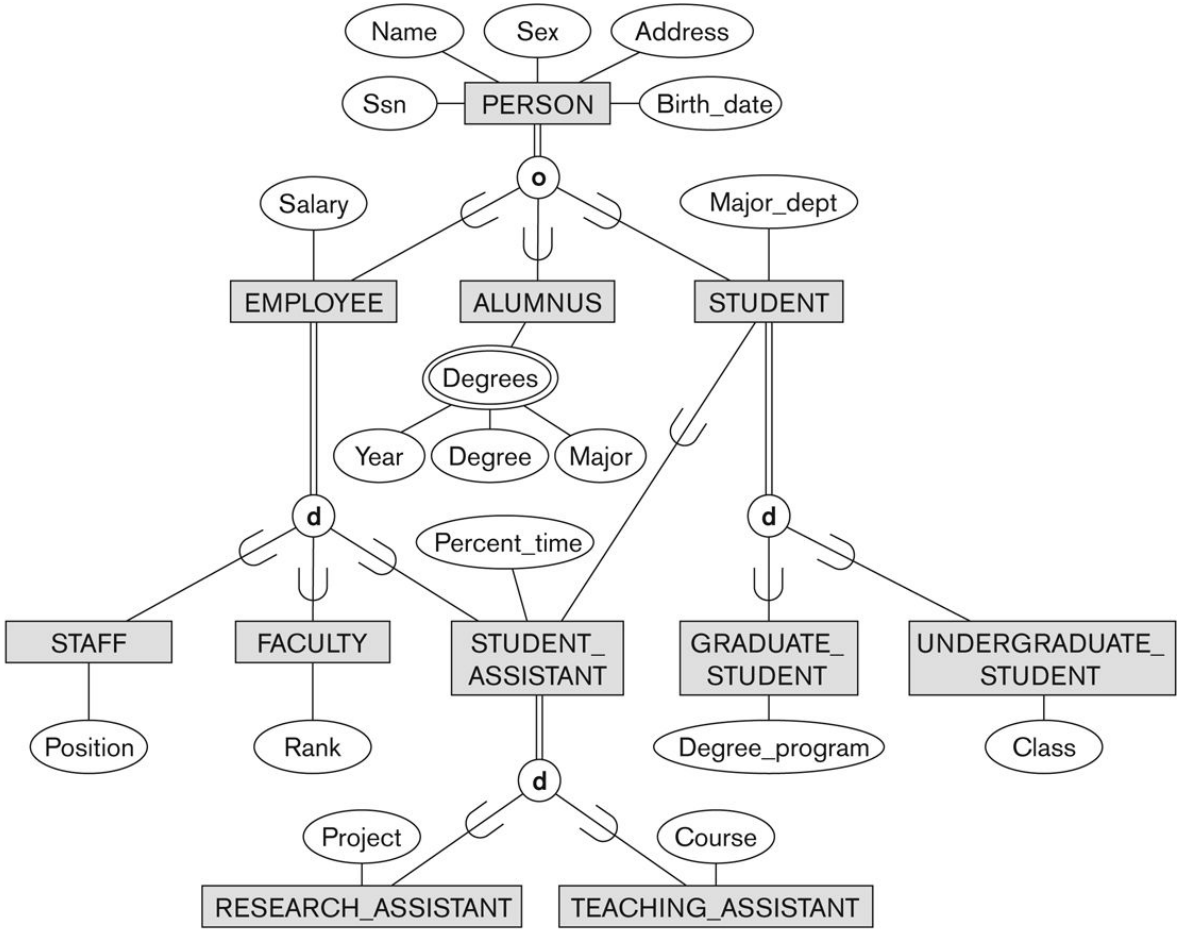

Figure 4.7
A specialization lattice with multiple inheritance for a UNIVERSITY database.

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Categories (UNION TYPES) (1)

- All of the *superclass/subclass relationships* we have seen thus far have a single superclass
- A shared subclass is a subclass in:
  - *more than one* distinct superclass/subclass relationships
  - each relationships has a single superclass
  - shared subclass leads to multiple inheritance
- **A Category (also called Union type or Union subclass) is a special kind of subclass that is formed from two or more DIFFERENT superclasses.**

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Categories (UNION TYPES) (2)

- Example: In a database for vehicle registration, a vehicle owner can be a PERSON, a BANK (holding a lien on a vehicle) or a COMPANY.
  - A *category* (UNION type) called OWNER is created to represent a subset of the *union* of the three superclasses COMPANY, BANK, and PERSON
  - A category member must exist in ***at least one*** of its superclasses
- Difference from *shared subclass*, which is a:
  - subset of the *intersection* of its superclasses
  - shared subclass member must exist in ***all*** of its superclasses
- The specialization is neither disjoint nor overlapping—it's a union.

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Two categories (UNION types): OWNER, REGISTERED_VEHICLE

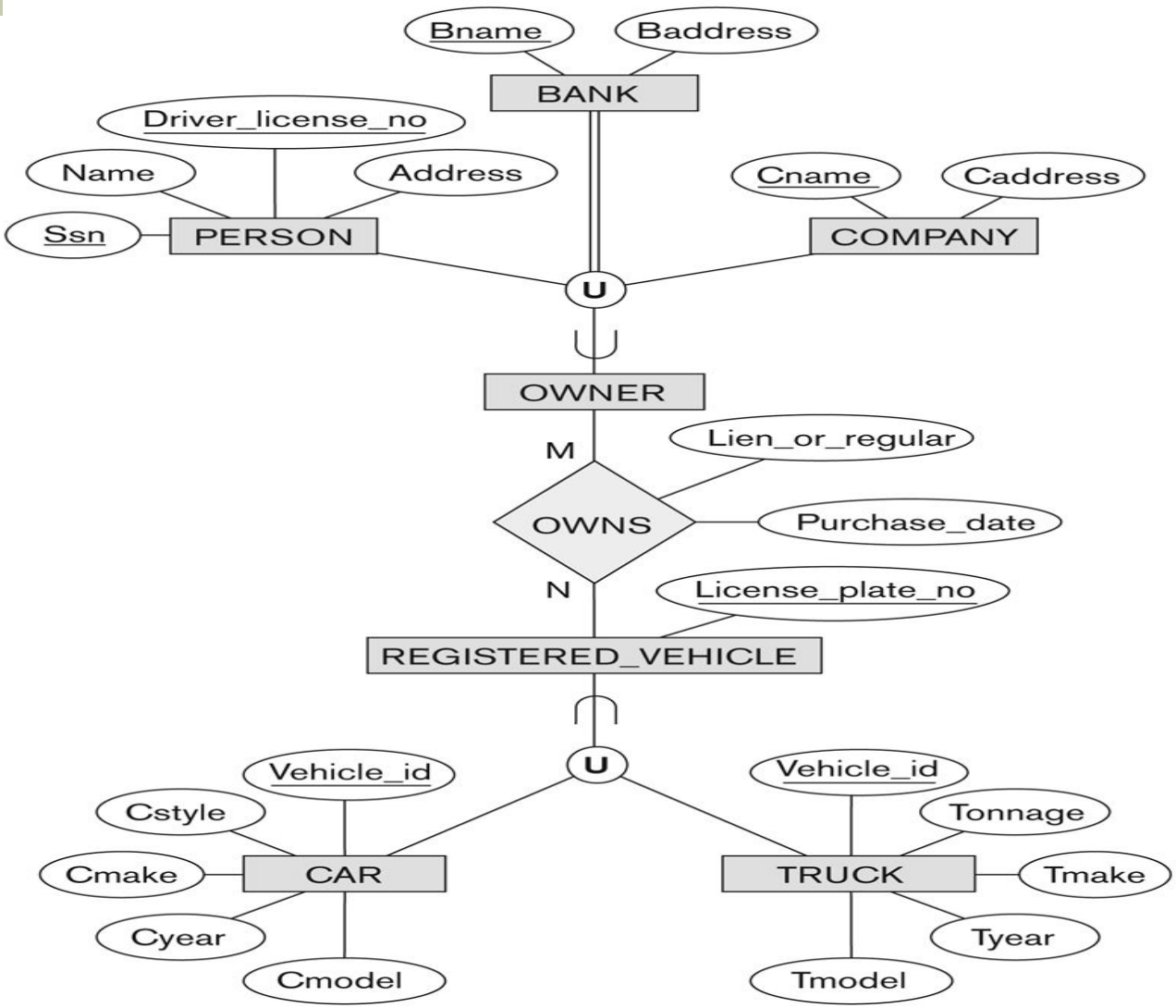

Figure 4.8
Two categories (union types): OWNER and REGISTERED_VEHICLE.

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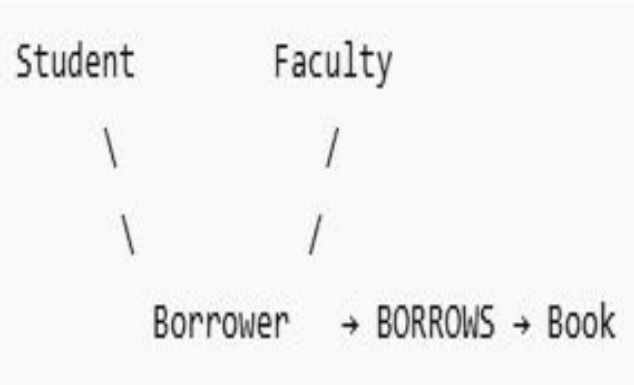
Example

- Borrower in Library System-
- Superclasses: Student & Faculty
- Category (Union subclass): Borrower
- relationship:
  - Borrower BORROWS Book
- A book can be borrowed by either a Student OR a Faculty member.
- Instead of creating two separate relationships:
  - Student–Borrows–Book
  - Faculty–Borrows–Book
- We create one common category: Borrower.



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Formal Definitions of EER Model (1)

- Class C:
  - A type of entity with a corresponding set of entities:
    - could be entity type, subclass, superclass, or category
- Note: The definition of *relationship type* in ER/EER should have 'entity type' replaced with 'class‘ to allow relationships among classes in general
- Subclass S is a class whose:
    - Type inherits all the attributes and relationship of a class C
    - Set of entities must always be a subset of the set of entities of the other class C
      - $S \subseteq C$
    - C is called the superclass of S
    - A superclass/subclass relationship exists between S and C

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Formal Definitions of EER Model (2)

- Specialization Z: Z = {S1, S2,…, Sn} is a set of subclasses with same superclass G; hence, G/Si is a superclass relationship for i = 1, …., n.
  - G is called a generalization of the subclasses {S1, S2,…, Sn}
  - Z is total if we always have:
    - $S1 \cup S2 \cup \ldots \cup Sn = G$;
    - Otherwise, Z is partial.
  - Z is disjoint if we always have:
    - $Si \cap Sj$ empty-set for $i \neq j$;
  - Otherwise, Z is overlapping.

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Formal Definitions of EER Model (3)

- A subclass not defined by a predicate is called **user-defined .**
- **User-defined specialization** is a type of specialization in which the membership of an entity in a subclass is determined by a **rule or condition** specified by the **user**.
- **Superclass (G):**
  - The higher-level entity set that is being specialized into multiple subclasses.
- **Subclasses (Si):**
  - The lower-level entity sets that represent specialized versions of the superclass.
- **User-Defined Rules**:
  - The conditions used to define membership in each subclass.
  - These rules can be complex and involve multiple attributes, derived attributes, or external computations.
- **Disjoint or Overlapping:**

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Example

Superclass: Employee

- Attributes: **YearsOfExperience**, **Salary**, **Role**.
- Subclasses and Conditions:
  - S_1: **Senior Employee**
    - Condition: $YearsOfExperience > 10$ AND $Salary > 100,000$.
  - S_2: **Manager**
    - Condition: $Role = "Manager"$.
  - S_3: **Part-Time Employee**
    - Condition: $Role = "Part-Time"$.

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Attribute-defined specialization

- **Superclass (G):**
  - The higher-level entity set that is being specialized into multiple subclasses.
- **Subclasses (Si):**
  - The lower-level entity sets that represent specialized versions of the superclass.
- **Defining Attribute (A):**
  - A single attribute of the superclass G whose value determines the membership of an entity in a particular subclass.
- **Predicate (A=ci):**
  - A condition or rule used to define membership in each subclass,
  - where ci is a constant value from the domain of the attribute A.

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- **Example:** A="$Teacher$", A="$Doctor$".
- Disjoint Condition (if applicable):
- If $ci \neq cj$ for $i \neq j$, and
- A is a single-valued attribute, the specialization is disjoint, meaning an entity can belong to only one subclass.

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Alternative diagrammatic notations

- ER/EER diagrams are a specific notation for displaying the concepts of the model diagrammatically
- DB design tools use many alternative notations for the same or similar concepts
- One popular alternative notation uses *UML class diagrams*
- see next slides for UML class diagrams and other alternative notations

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UML Example for Displaying Specialization / Generalization

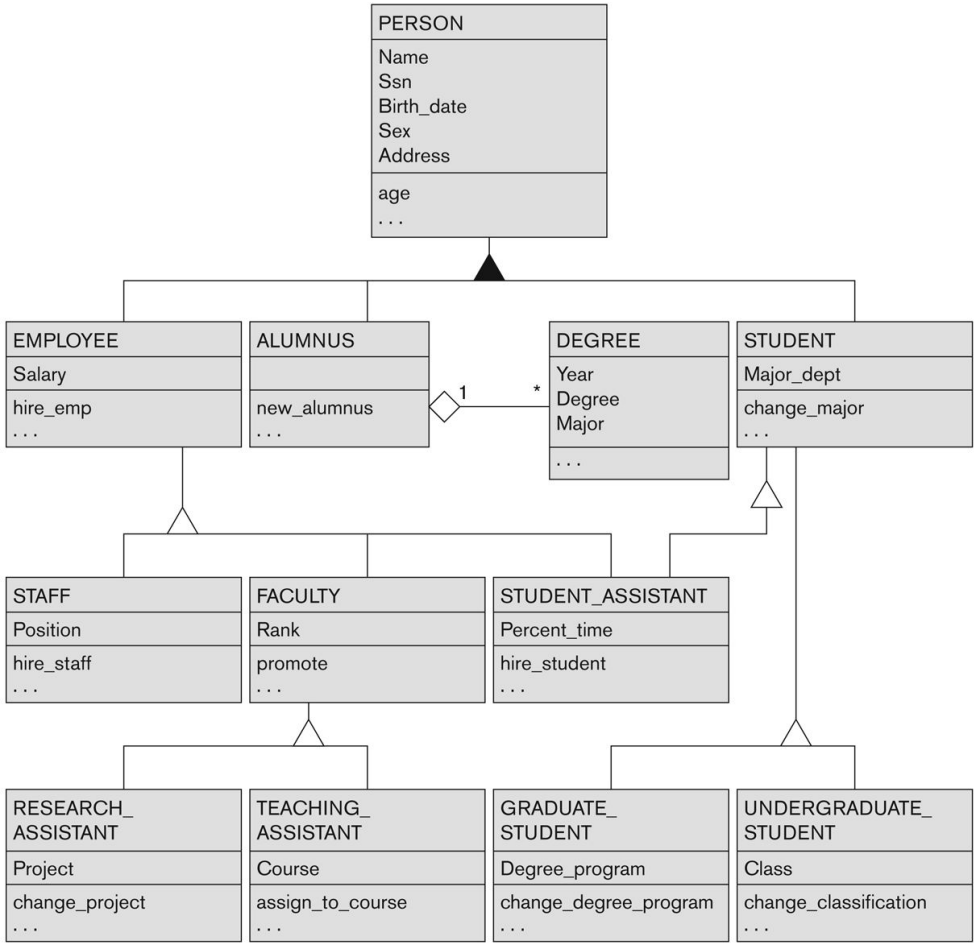

Figure 4.10
A UML class diagram corresponding to the EER diagram in Figure 4.7, illustrating UML notation for specialization/generalization.

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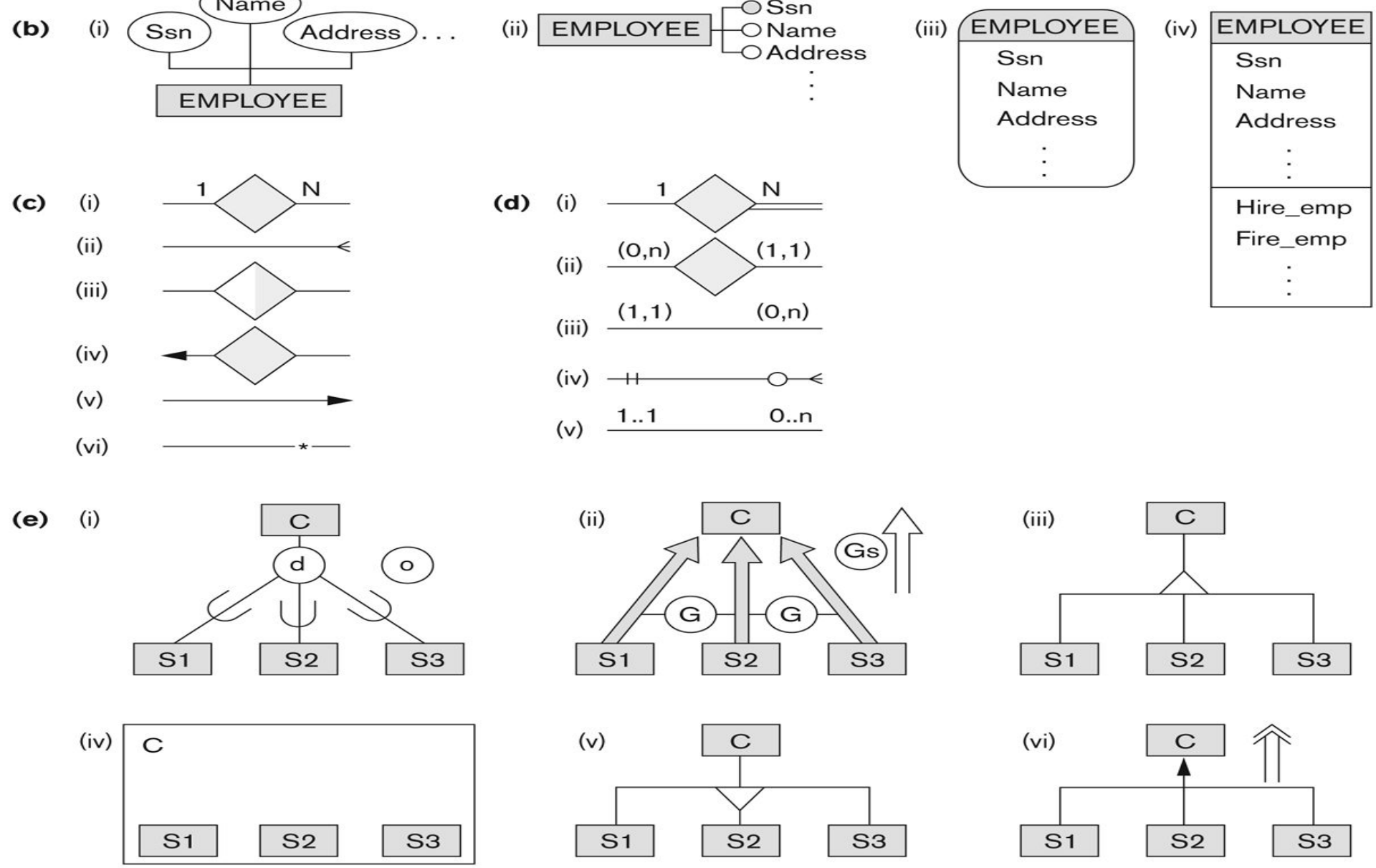

Figure A.1
Alternative notations. (a) Symbols for entity type/class, attribute, and relationship. (b) Displaying attributes. (c) Displaying cardinality ratios. (d) Various (min, max) notations. (e) Notations for displaying specialization/generalization.

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General Conceptual Modeling Concepts

- GENERAL DATA ABSTRACTIONS
  - CLASSIFICATION and INSTANTIATION
  - AGGREGATION and ASSOCIATION (relationships)
  - GENERALIZATION and SPECIALIZATION
  - IDENTIFICATION
- CONSTRAINTS
  - CARDINALITY (Min and Max)
  - COVERAGE (Total vs. Partial, and Exclusive (disjoint) vs. Overlapping)

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Ontologies

- Use conceptual modeling and other tools to develop "a specification of a conceptualization"
  - **Specification** refers to the language and vocabulary (data model concepts) used
  - **Conceptualization** refers to the description (schema) of the concepts of a particular field of knowledge and the relationships among these concepts
- Many medical, scientific, and engineering ontologies are being developed as a means of standardizing concepts and terminology

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Summary

- Introduced the EER model concepts
  - Class/subclass relationships
  - Specialization and generalization
  - Inheritance
- EER diagrams and alternative notations were presented

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